People and Processes



Year 9 Computing Technology

Learning Intention



Explore careers in software development

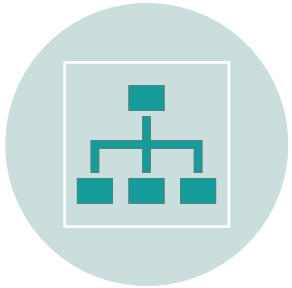


Explore interests and careers in analysing data

Success Criteria



I can describe a broad range of roles in computing



I can compare users, developers, operators and management.

People and Processes in Information Systems

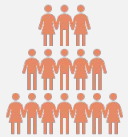


Information systems rely on **people and processes**



Two main groups are involved:

Users
Technical experts



People are involved in:

Development
Operation
Management



IT careers exist across many industries:

Health, education, defence, agriculture, engineering, entertainment

Users

User: an entity—typically a person, but sometimes a machine or automated service—that interacts with a system, application, or website to perform tasks, access information, or consume services



Users are the people who **use the system daily**

Successful systems are designed with **user needs in mind**

Systems should be:

- Easy to use
- Logical
- Intuitive

User processes are supported by:

- Clear interfaces
- Documentation (e.g. user manuals)

Example user tasks:

- Logging in
- Uploading files
- Searching
- Navigating menus

Developers

Developers design and build information systems

They turn user needs into **working software**

Common developer roles include:

- Computer scientists
- Programmers
- Software engineers

These roles usually require **tertiary study**

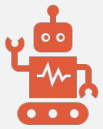
Developers focus on:

- Coding
- Testing
- Problem-solving
- Improving systems

Computer Scientists



Design new hardware and software technologies



Work in areas such as:

- Security
- Robotics
- AI
- Medical research



Require strong theoretical knowledge



Typically have computer science or engineering degrees

Programmers and Software Engineers

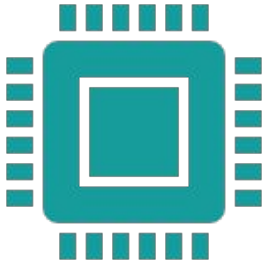


Programmers:

Write and maintain code

Test and debug software

Translate solutions into programming languages



Software engineers:

Design large software systems

Develop operating systems and applications

Fix and improve existing software

Emerging IT Roles

AI / Machine Learning Engineers: build intelligent systems

Data Scientists: analyse large data sets for insights

Cybersecurity Analysts: protect systems and data

Cloud Architects: design cloud-based systems

Blockchain Developers: build decentralised systems

IoT Architects: manage connected smart devices

VR/AR Developers: create immersive digital experiences

Quantum Computing Engineers: work with advanced computing technology

Operators and Administrators



Ensure systems run **reliably and efficiently**



Work behind the scenes



Key roles include:

- Computer operators
- Database administrators



Focus on maintenance, updates and monitoring

Computer Operators



Keep critical systems running (e.g. servers)



Responsibilities include:

- Updating software
- Monitoring storage and memory
- Checking system environments
- Identifying faults



Often work with specialised system software

Database Administrators



Create and maintain databases



Ensure data is:

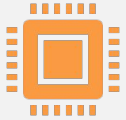
- Secure
- Accurate
- Available



Work with:

- Programmers
- Systems analysts
- Support many business processes

Systems Analysts



Bridge the gap between users and developers



Analyse problems and design solutions



Specialise in systems such as:

- Financial
- Accounting
- Engineering



Focus on improving or upgrading systems

IT Support Staff



Help users solve technical problems



Work in help desks or support teams



Tasks include:

- Troubleshooting errors
- Fixing hardware issues
- Interpreting error messages



More complex repairs handled by technicians

Training Specialists

Teach people how to use technology

Plan and deliver IT training

Support:

- Workers
- Students
- Organisations

Important as technology changes rapidly

Management Roles



- Focus on planning and decision-making
- Ensure information systems support business goals
- Key management roles include:
  - Chief Information Officer (CIO)
  - Project managers
  - Information security officers

Chief Information Officer (CIO)



Senior management role



Oversees information systems strategy



Responsibilities include:

- Budgeting
- Scheduling
- Aligning IT with business goals



Strong management and communication skills required

Project Managers



Manage IT projects from start to finish



Ensure projects:

Stay on schedule
Stay within budget



Coordinate teams and tasks



Used across many IT projects (apps, games, networks)

Information Security Officer



Protects systems and data



Manages security policies and procedures



Prevents:

- Hacking
- Viruses
- Data breaches



Cybersecurity is a rapidly growing field